Samson Niyizurugero

Machine Learning Engineer & AI Specialist

Available for: Part-time/Project-based (Immediate) | Remote | 12-20 hours/week
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LinkedIn | GitHub | Hugging Face | My Website

Professional Profile

Machine Learning Engineer with an MSc in Mathematical Sciences from AIMS Rwanda (2023) and 2+ years experience building and deploying production ML models. Specialized in LLM-powered agents (LangChain, RAG), recommendation systems, predictive analytics, and model monitoring. Built 8+ live deployed AI applications on Hugging Face Spaces with end-to-end ML pipelines. Strong fundamentals in TensorFlow, PyTorch, vector databases, embeddings, and MLOps. Seeking to build Africa's first AI-powered wellness platform with scalable, production-ready intelligent systems.

Production ML & AI Applications (Click to View Live Demos)

FarmSmart Agentic AI: Live Demo – LLM-powered agricultural advisor using LangChain, RAG, tool-use agents, and real-time APIs. Production-deployed on Hugging Face Spaces.

Informal Address Resolver: Live Demo – Production geocoding system optimizing for <100ms latency, offline capable, 94% within 300m.

Rwanda Air Quality Dashboard: Live Demo – LSTM-based forecasting with health impact analysis. Presented at SOSCHI 2025.

Tourist Guide AI Agent: Live Demo – Agentic AI with chain-of-thought reasoning and tool-use capabilities for personalized recommendations.

AI Math Tutor: Live Demo – Adaptive multilingual tutor with offline capability. Built for AIMS KTT Hackathon.

Credit Risk Assessment: GitHub – XGBoost with SHAP interpretability for financial ML.

CPI Tender Matcher: Live Demo – Hybrid TF-IDF recommendation engine for grant matching.

African CO2 Emissions Platform: Live Demo – ML forecasting across 54 African countries.

Professional Experience

AIMS Research and Innovation Centre (AIMSRIC), Kigali **2023–2025**
ML Engineer and Data Manager

- Built and deployed 6+ ML models to production (Hugging Face Spaces) with automated monitoring for prediction drift and accuracy degradation
- Developed production-ready agentic AI systems (LangChain, RAG) deployed as public-facing applications serving thousands of users
- Implemented end-to-end ML pipelines including data preprocessing, model training, evaluation, and API deployment using FastAPI
- Engineered vector database solutions for RAG systems, optimizing retrieval latency and response quality
- Managed data infrastructure for 120 IoT monitoring nodes including ETL pipelines, quality assurance, and PostgreSQL databases
- Deployed LSTM-based time series forecasting models achieving 23% error reduction compared to baseline
- Presented research at MLOA 2025 and SOSCHI Conference on Climate-Health Interactions

Willows International, Kigali **2021–2022**
Data Analyst

- Built automated reporting pipelines and dashboards using Power BI and Python
- Conducted statistical analysis and predictive modeling for program evaluation
- Designed data collection systems and survey instruments for field deployment

Technical Skills

ML & Deep Learning
TensorFlow & PyTorch
Scikit-learn & XGBoost
LSTM & Time Series
Computer Vision (CNN)
Ensemble Methods
Model Evaluation & Tuning

LLMs & Agentic AI
LangChain & LangGraph
RAG Systems
Vector Databases (Chroma, FAISS)
Embeddings & Semantic Search
Prompt Engineering
Chain-of-Thought Reasoning
Tool-Use & Function Calling

MLOps & Deployment
Hugging Face Spaces
FastAPI & Flask
Docker & Containerization
CI/CD Pipelines
Model Monitoring & Drift Detection
Cloud (AWS/GCP)

Data Engineering
Python (pandas, numpy)
SQL (PostgreSQL, MySQL)
ETL/ELT Pipelines
API Development
Data Quality Assurance

Programming
Python & R
SQL
JavaScript
Swift (iOS)
Bash & Git

Recommendation Systems
Collaborative Filtering
Content-Based Filtering
Hybrid Systems
TF-IDF & Embeddings
Personalization Algorithms

Core Competencies

Machine Learning & AI: Supervised/Unsupervised Learning, Deep Learning (TensorFlow, PyTorch), Time Series (LSTM), NLP, LLMs, Agentic AI (LangChain), RAG, XGBoost, Random Forest, Ensemble Methods
Model Interpretability: SHAP, LIME, Feature Importance, Explainable AI (XAI)
Production ML: Model Deployment, API Development, Model Monitoring, Drift Detection, Performance Optimization
Recommendation Engines: Collaborative Filtering, Content-Based Filtering, Hybrid Systems, Vector Similarity
Geospatial ML: Spatial-Temporal Analysis, Remote Sensing, GIS (QGIS, PostGIS)
Healthcare & Wellness: Biostatistics, Global Health ML, Bioacoustics, Fitness Analytics, Habit Modeling

Detailed AI & Machine Learning Projects

- **FarmSmart Agentic AI – Production LLM System** [Live Demo]
  Built production AI agent using LangChain and RAG providing personalized recommendations. Integrates vector database for soil knowledge, real-time weather APIs via tool-use, and chain-of-thought reasoning. Deployed on Hugging Face Spaces with monitoring.
- **Tourist Guide AI Agent – Agentic AI with Tool Use** [Live Demo]
  Implemented agentic AI system with chain-of-thought reasoning, multi-tool orchestration (weather APIs, maps, recommendations), and conversational memory. Demonstrates production LLM agent architecture.
- **Credit Risk Assessment – XGBoost with SHAP** [GitHub]
  Built gradient boosting model achieving 87% accuracy. Implemented SHAP for full model interpretability. Demonstrates production-ready financial ML with explainable AI.
- **Informal Address Resolver – Production Geocoding** [Live Demo]
  Engineered low-latency geoparsing system (<100ms, offline-capable). Includes comprehensive test suite and performance monitoring.
- **Rwanda Air Quality LSTM Forecasting** [Live Demo]
  Developed LSTM-based time series models with hyperparameter optimization and drift monitoring. Deployed as real-time dashboard with health impact integration.
- **AI Math Tutor – Production Educational AI** [Live Demo]
  Built adaptive multilingual tutor with offline capability, speech recognition, and difficulty scaling. Demonstrates practical AI deployment for real users.

Education

African Institute for Mathematical Sciences (AIMS), Rwanda **2021–2023**
Master of Science in Mathematical Sciences (Major: Data Science) Graduated: June 2023
Thesis: **Machine Learning and Time Series Analysis for Environmental Early Warning Systems** [View Thesis]

University of Rwanda **2018–2021**
Bachelor of Science in Statistics
Coursework: Biostatistics, Survival Analysis, Econometrics, Design of Experiments, Time Series Forecasting

Certifications & Professional Training

- **AIMS RIC Doctoral Training School 2026** – Foundational Methods in Data Science [Certificate]
- **East-African School on Advanced Machine Learning (EASY-ML)** – CNRS, Sorbonne, University of Nairobi, Kenya (2025)
- **Synthetic Data for Healthcare Research** – Washington University St. Louis, DS-I Africa, AIMS Rwanda, NIH (2023)
- **Modern Statistics & Machine Learning for Global Health** – Imperial College London and AIMS (2024)
- **Deep Learning for Wildlife Monitoring (Bioacoustics & Biologging)** – AIMS RIC (2025)
- **Building Your Own Chatbot (RAG, Fine-tuning LLMs, Deployment)** – AIMS RIC (2025)
- **Optimization, Optimal & Predictive Control Workshop** – AIMS Rwanda and AIMS RIC (2025)
- **Geospatial Analytics using R and R-INLA** – MRC, Imperial College (2023)
- **Monitoring & Evaluation for Development Programs** – World Bank Group (2024)
- **Results-Based Management & Impact Evaluation** – UNDP (2023)
- **ICDL Data Analytics Certification** – ICDL Africa (2024)
- **Certificate in Business Management** – ESMT Berlin, iiAfrica, Harvard Business Publishing (2023)
- **App Development with Swift** – Apple Authorised Training (2025)
- **Gender Equality and Global Markets** – IFC World Bank, UN Women, Columbia SIPA (2024)
- **African Mathematical School & Conference** – INES-Ruhengeri, AMU, AIMS Rwanda (2023)
- **Summer School A-PIC 2025** – Justus-Liebig-Universität Giessen, AIMS Rwanda (DAAD)

[All certificates and degrees: Click to view all Certificates]

Research Publications and Conference Presentations

- **Co-author:** *City-wide space-time patterns of environmental noise pollution in Kigali, Rwanda*
  Environmental Research Letters, 2024 [View Paper]
- **Certificate of Recognition and Poster Presentation:** *Development and Validation of a Machine Learning Based Early Warning System for Air Pollution in Rwanda*
  Second African Conference on Machine Learning, Optimization and Applications (MLOA 2025), AIMS and TU Ilmenau, September 2025
- **Research Presenter:** *Early Warning System for Air Pollution in Rwanda*
  SOSCHI Conference on Standards for Official Statistics on Climate-Health Interactions, Kigali, Rwanda, December 2025 [View Certificate]
- **In Preparation:** *Modeling Adoption Patterns of Improved Cookstoves: A Multi-stage Analysis of Household Energy Transitions in Rwanda's Northern Province*

Awards and Recognition

- Certificate of Recognition – Second African Conference on Machine Learning, Optimization and Applications (MLOA 2025), AIMS and TU Ilmenau
- Research Presenter – SOSCHI Conference on Standards for Official Statistics on Climate-Health Interactions, Kigali (2025)
- AIMS KTT Hackathon Participant – Data Engineering Track, Agentic AI Track, Geospatial Track (2025–2026)
- AIMS Graduate – African Institute for Mathematical Sciences (2023)
- Summer School Participant – A-PIC 2025, Justus-Liebig-Universität Giessen and AIMS Rwanda (DAAD)

Research and Professional Interests

Production ML systems and MLOps; LLM-powered agents and RAG architectures; Recommendation systems for wellness and fitness; Model interpretability (SHAP, LIME); Deep learning for time series; AI for health, wellness, and sport performance.

AIMS Community Engagement

- AIMS MSc Graduate (2023 Cohort)
- AIMS KTT Hackathon Participant – Data Engineering, Agentic AI, Geospatial Tracks (2025–2026)
- AIMS RIC Doctoral Training School Participant (2026)
- Summer School A-PIC 2025 – Justus-Liebig-Universität Giessen and AIMS Rwanda (DAAD)
- African Mathematical School Participant – INES-Ruhengeri (2023)

References

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